

Model: Advancement under Local Competition

We consider individuals embedded in local comparison groups (“pools”), such as evaluation cohorts, teams, or departments. Each individual i has baseline ability x_i and belongs to a pool p with quality q_p . Time spent in the pool affects performance through a monotone benefit:

$$s_i = x_i + b(q_p) + \varepsilon_i,$$

where $b'(q) > 0$ captures developmental or **signaling gains** from stronger environments, and ε_i is idiosyncratic **noise**.

Advancement is determined by relative standing within the pool. Each pool has n_p individuals competing for k_p advancement slots. Individuals are promoted if their score exceeds the **endogenous cutoff** for the top k_p :

$$s_i \geq T_p(q_p)$$

where $T_p(q_p)$ is the $\left(1 - \frac{k_p}{n_p}\right)$ - quantile of the pool's score distribution.

The probability of advancement for an individual with ability $x_i = x$ is therefore:

$$P(\text{advance}_i = 1 \mid x, q_p) = P(x + b(q_p) + \varepsilon_i \geq T_p(q_p))$$

This formulation contains two opposing, monotone forces:

- **Development:** higher pool quality increases individual performance through $b(q_p)$.
- **Competition:** higher pool quality raises the cutoff $T_p(q_p)$, because peers are stronger and slots are scarce.

The relationship between **pool quality** and **advancement** depends on the difference between these forces. Differentiating the advancement probability shows that its slope is governed by:

$$b'(q_p) - T'_p(q_p)$$

If developmental gains dominate at low q_p and competitive pressure dominates at high q_p , then there exists an **interior point** q^* such that:

$$b'(q^*) = T'_p(q^*)$$

and advancement probability is **increasing** in q_p for $q_p < q^*$, and **decreasing** for $q_p > q^*$.

This produces an inverted U-shaped relationship between pool quality and individual success.

Importantly, the non-monotonicity is not imposed. It emerges from

- i. monotone improvements in performance with environment quality and

- ii. rank-based selection with limited advancement opportunities.

Interpretation

This model treats advancement as a contextual, relational process: individuals are shaped by their environment, but evaluated against it. Stronger environments simultaneously enhance capability and intensify competition. As a result, improving the quality of one's local comparison group yields diminishing and eventually negative returns for individual advancement.

Other Stuff

success in competitive systems depends on relative position within local groups, leading to non-monotonic returns to environment quality,

this arises from the interaction of performance gains and rising selection thresholds

Advancement in competitive systems is governed by a fundamental tradeoff between developmental gains from stronger environments and congestion from relative competition under scarce opportunities.

The right model would give us a strong “counterfactual lever”:

You need a clear implication like:

~ expanding slots increases mobility but changes where the peak sits

~ concentrating talent (sorting) can reduce individual advancement

~ policies that create elite clusters may reduce mobility within them

==> stronger environments are not unambiguously beneficial

Network angle

these are structured local comparison systems, and a network representation provides a common formal language for them.

“does the network representation add anything beyond the pool variable?” If the answer is only that it unifies the cases and makes the model more general, that is still useful, just not central. If the answer is that it changes the empirically relevant comparison set, then it becomes much more powerful.

Networks beyond pools = do we get better models by taking the union over the rater pools / teams when kids transfer than taking just the "last" rater pool?